MARYAM REHAN

INTERNSHIP WEEK 01 TASK 03 CYBER SECURITY (SOC)

SOC Lab Setup and Preparation Report

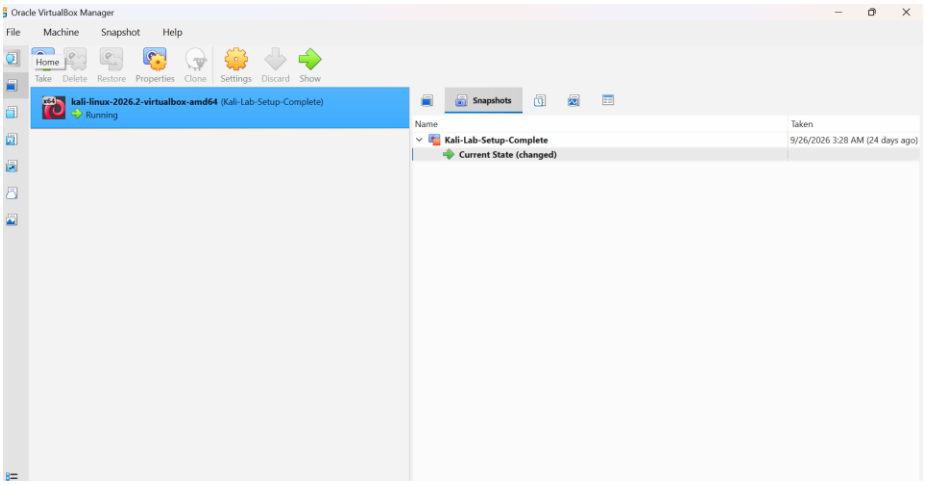
1. Objective

The objective of this task is to prepare a virtual lab environment using Oracle VirtualBox, Kali Linux, and Windows for basic SOC operations, security monitoring, and network connectivity testing.

2. Tools Used

- Oracle VirtualBox
- Kali Linux 2026.2
- Linux Terminal
- VirtualBox Network Adapter

Screenshot 01: VirtualBox Manager



Description:

Oracle VM VirtualBox Manager was used to manage the Kali Linux virtual machine. The interface displays the available virtual machines and their current status. This helps in starting, stopping, and managing the virtual environment used for Linux practical exercises.

Purpose:

To demonstrate the virtual machine management environment.

Screenshot 02: Kali Linux Running



Description:

Kali Linux was successfully started inside Oracle VirtualBox. The desktop environment provides access to Linux utilities and the terminal used for practical exercises.

Purpose:

To demonstrate that the Kali Linux virtual machine is running successfully.

<u>Demo of 10 Linux commands:</u>

```
┌──(root㉿kali)-[~]
└─# pwd
/root
```

Displays the current working directory

```
┌──(root㉿kali)-[~]
└─# ls
cookies.txt  Downloads  Mediroza  Module1
```

Current directory files folders Listing

```
┌──(root㉿kali)-[~]
└─# whoami
root
```

Current logged-in username show

```
┌──(root㉿kali)-[~]
└─# date
Thu Oct 15 12:54:15 PM EDT 2026
```

System date time show

```
┌──(root㉿kali)-[~]
└─# mkdir demo
```

Demo folder create

```
┌──(root㉿kali)-[~]
└─# cd demo

┌──(root㉿kali)-[~/demo]
└─#
```

Inside Demo folder

```
┌──(root㉿kali)-[~/demo]
└─# touch file.txt
```

Empty text file create

```
┌──(root㉿kali)-[~/demo]
└─# echo "Hello Linux" > file.txt
```

Write text inside file

```
┌──(root㉿kali)-[~/demo]
└─# cat file.txt
Hello Linux
```

Display Text inside file

```
┌──(root㉿kali)-[~/demo]
└─# cp file.txt copy.txt
```

Create Copy of file

Detailed Description

1. Display the Current Working Directory

Command: pwd

Description: This command displays the full path of the current working directory in the Linux terminal.

2. List Files and Folders

Command: ls

Description: This command lists the files and folders available in the current directory.

3. Display the Current Logged-in Username

Command: whoami

Description: This command displays the username of the currently logged-in user.

4. Display System Date and Time

Command: date

Description: This command displays the current system date and time.

5. Create a Demo Folder

Command: mkdir Demo

Description: This command creates a new directory named Demo in the current working directory.

6. Navigate Inside the Demo Folder

Command: cd Demo

Description: This command changes the current working directory to the newly created Demo folder.

7. Create an Empty Text File

Command: touch file.txt

Description: This command creates an empty text file named file.txt inside the current directory.

8. Write Text Inside the File

Command: echo "Hello Linux" > file.txt

Description: This command writes the text “Hello Linux” into file.txt. The > operator redirects the text into the file and overwrites existing content.

9. Display Text Inside the File

Command: cat file.txt

Description: This command displays the contents of file.txt in the terminal.

10. Create a Copy of the File

Command: cp file.txt copy.txt

Description: This command creates a copy of file.txt named copy.txt in the same directory.

Result

The practical demonstrated ten basic Linux commands for directory navigation, file management, user identification, and displaying system information. These commands provide a foundation for working with the Linux command-line interface.

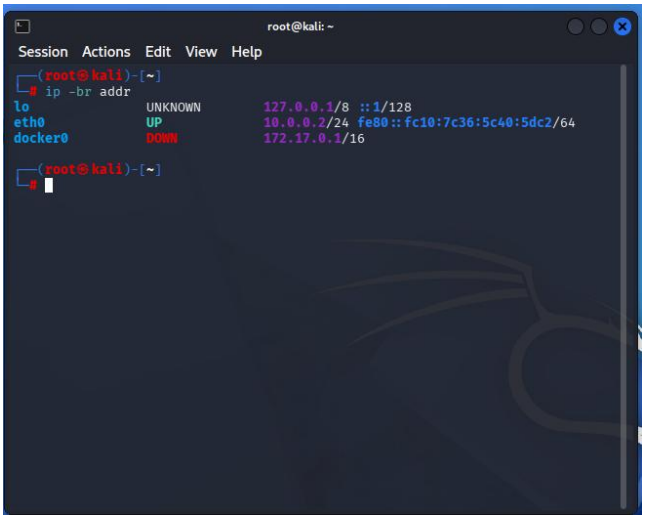
Description:

Kali Linux was started successfully inside Oracle VM VirtualBox. The desktop environment provides access to Linux utilities, applications, and the terminal. This environment was used to perform basic Linux commands and networking exercises.

Purpose:

To demonstrate the Kali Linux operating environment.

Network Information



Command Used:

ip -br addr

Description:

The ip -br addr command was executed to display network interfaces and their IP addresses in a brief format.

The output showed the following interfaces:

- **lo:** The loopback interface with IP address 127.0.0.1/8, used for local communication.
- **eth0:** The network interface was UP with IPv4 address 10.0.0.2/24 and an IPv6 link-local address.
- **docker0:** The Docker virtual network bridge with IP address 172.17.0.1/16, currently in the DOWN state.

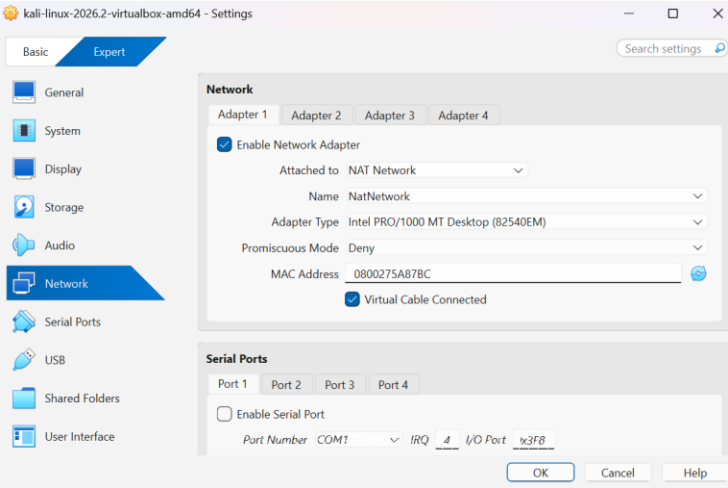
Result:

The command successfully displayed the network interface names, operational states, and assigned IP addresses.

Purpose:

To identify network interfaces and understand their current configuration.

Internal Network Configuration



Description:

The VirtualBox network settings allow virtual machines to communicate through an internal network. Both Kali Linux and Windows VMs must be configured with the same internal network name to enable communication between them.

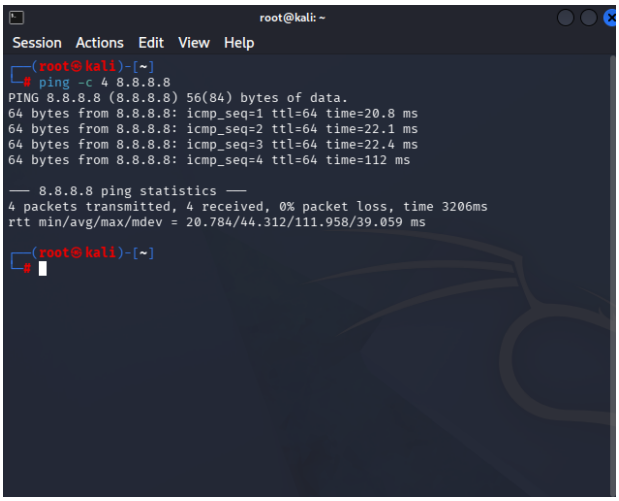
Purpose:

To demonstrate the network configuration required for communication between virtual machines.

Status:

Internal network configuration between Kali Linux and Windows is pending because the Windows virtual machine has not yet been set up. Communication between the two virtual machines has not been tested.

Network Connectivity Test



Command Used:

ping -c 4 8.8.8.8

Description:

The ping command was executed in the Kali Linux terminal to test connectivity to Google's public DNS server at 8.8.8.8. Four ICMP packets were transmitted, and all four packets were received successfully. The test showed 0% packet loss, with an average response time of 44.312 milliseconds.

Result:

The test confirmed that Kali Linux had successful IP connectivity to the external destination during the test.

Purpose:

To verify basic network connectivity from the Kali Linux virtual machine.

Note:

This test verifies external network connectivity only. Connectivity between Kali Linux and a Windows virtual machine has not yet been tested.

Final Status

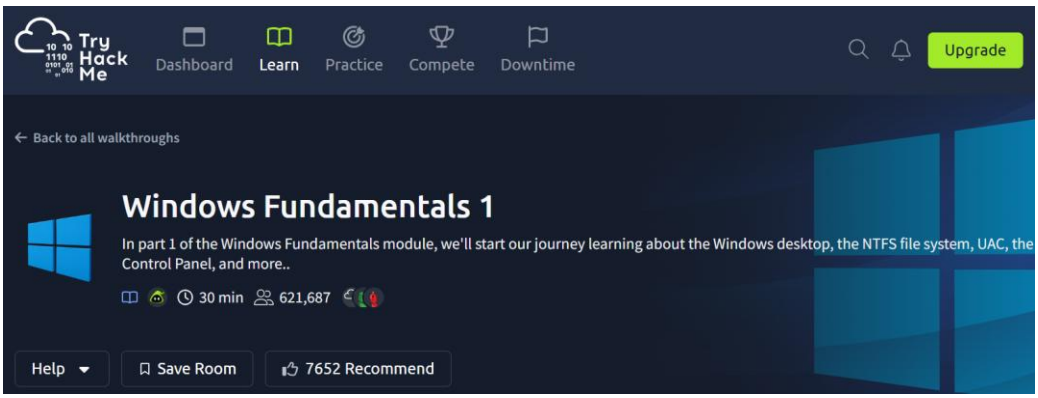
The initial SOC lab preparation was completed using Oracle VirtualBox and Kali Linux. The Kali Linux virtual machine was started successfully, ten basic Linux commands were practised, network interfaces were inspected, and external IP connectivity was verified with 0% packet loss.

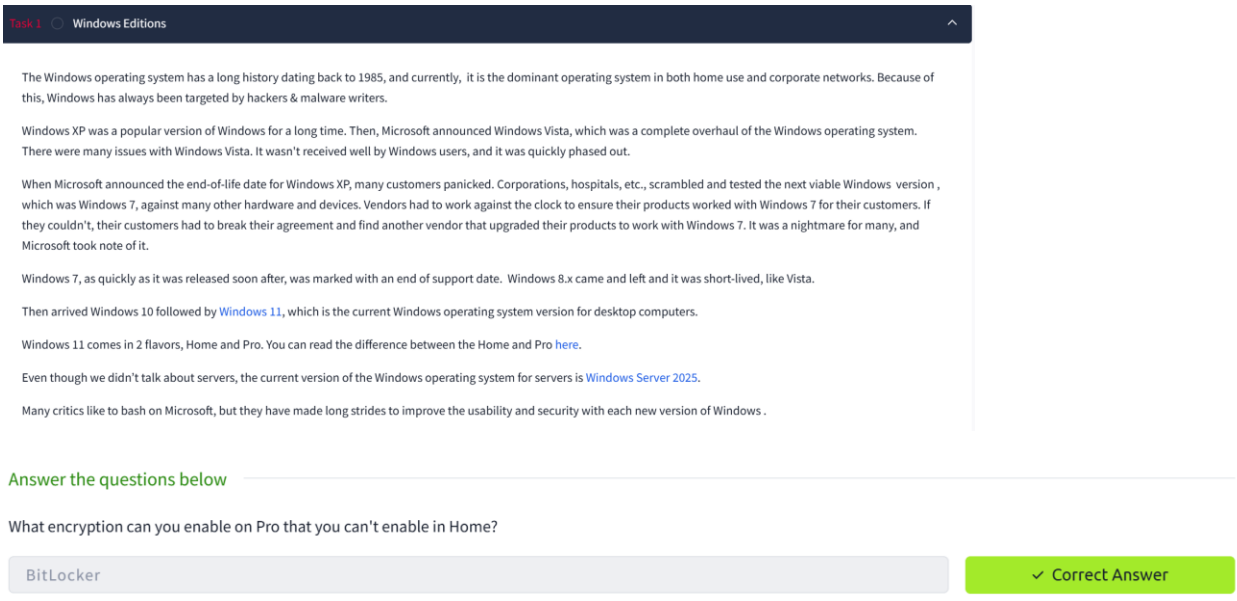
The Windows virtual machine setup and Kali-to-Windows connectivity test remain pending. Other configuration items, including shared folders and snapshots, have not been marked as completed because they have not yet been verified.

The Windows virtual machine has not yet been set up in VirtualBox, so communication between Kali Linux and Windows could not be tested. The shared folder, clipboard, drag-and-drop, and snapshot settings have not been verified because these configurations were not checked during the completed practical. Therefore, they have been left unconfirmed to ensure the report accurately reflects the work performed.

Conclusion

This practical provided initial experience with virtual machine management, Linux commands, and basic network troubleshooting. The completed work provides a foundation for further SOC lab activities. Additional setup and connectivity testing are required before the full two-machine SOC lab is ready.





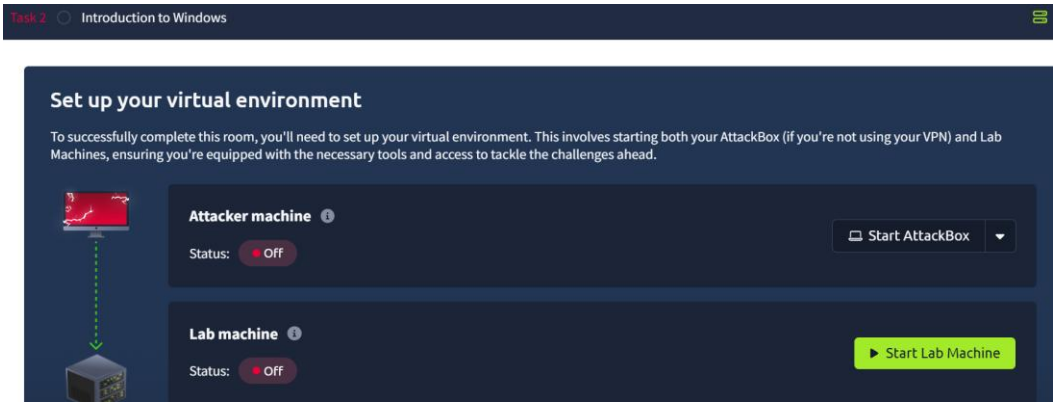
Description: This task explained the evolution of Microsoft Windows, from older versions such as Windows XP and Windows 7 to Windows 10 and Windows 11. It also introduced the differences between Windows Home and Pro editions.

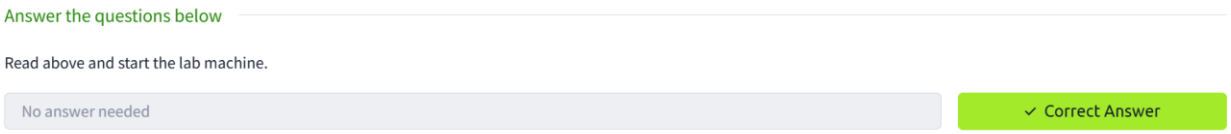
Answer: BitLocker

Learning Outcome: I learned that BitLocker Drive Encryption is a feature available in Windows Pro editions that provides drive encryption to help protect data from unauthorized access.

Result: The task question was answered by identifying BitLocker as the encryption feature available in Pro but not in Home.

Conclusion: In this task, I learned about the evolution of Microsoft Windows and the differences between Windows Home and Pro editions. I also learned that BitLocker Drive Encryption is available in Windows Pro to help protect data from unauthorized access. This task improved my understanding of Windows operating systems and basic security features.

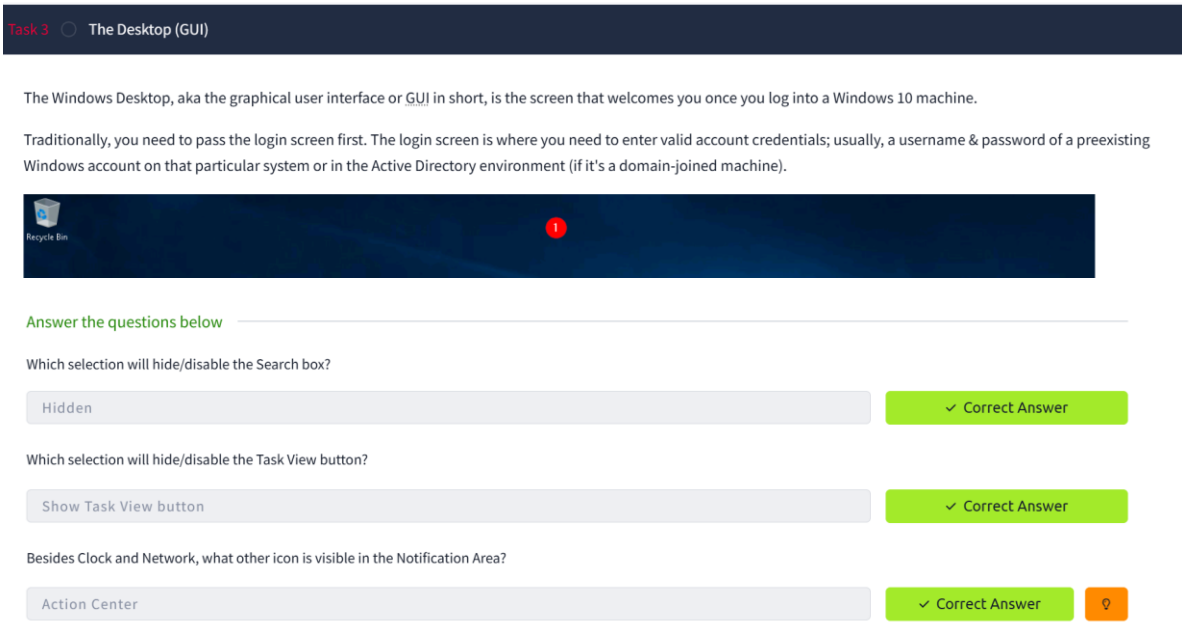




Description: This section explained how to set up the virtual environment required to complete the Windows Fundamentals 1 room on TryHackMe. It introduced two virtual machines: the AttackBox, which provides tools for cybersecurity activities, and the Lab Machine, which contains the Windows operating system for practical learning.The instructions explained how to start the Lab Machine and access it through a web browser. They also described how to connect remotely using Remote Desktop Protocol (RDP), with the provided machine IP address, username, and password. Users may need to accept the security certificate when prompted. The lab machine can take up to three minutes to become available.

Learning Outcome:I learned how to prepare the virtual lab environment and understood the basic process of accessing a Windows machine through a browser or Remote Desktop connection. Make the description consistent in tenseClarify the two access methods

Conclusion: This task provided an understanding of how to set up a virtual learning environment on TryHackMe. I learned about the purpose of the AttackBox and Windows Lab Machine, as well as the basic steps for accessing the lab through a web browser or Remote Desktop Protocol (RDP). This knowledge will help me perform Windows Fundamentals practical exercises in a controlled virtual environment.

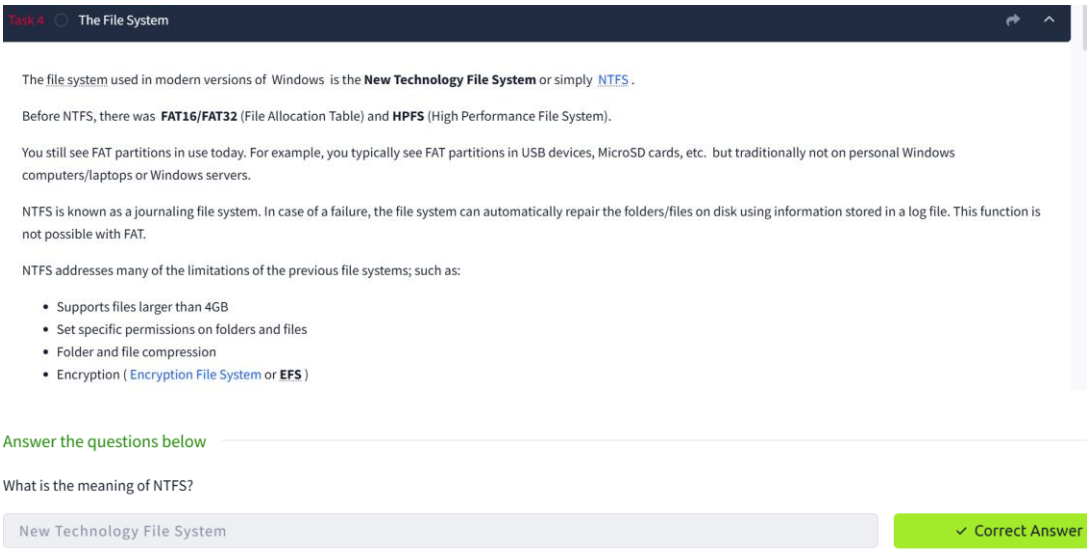


Description: This section introduced the Windows Desktop, also known as the Graphical User Interface (GUI), which appears after a user logs into Windows. It explained the main components of the desktop, including the Desktop, Start Menu, Search Box, Task View, Taskbar, Toolbars, and Notification Area. The section described how users can customize the desktop by changing the wallpaper, themes, display settings, and icon arrangements. It also explained how the Start Menu provides access to installed applications, settings, files, and power options. The Taskbar displays open or pinned applications, while the Notification Area shows the clock, network status, volume, and other system icons.

Learning Outcome:I learned about the main components of the Windows GUI and how to navigate the desktop, access applications, customize display settings, and manage taskbar and notification area options.

Conclusion: In this task, I learned about the Windows Desktop and its main graphical user interface components, including the Start Menu, Search Box, Task View, Taskbar, and Notification Area. I also learned how to customize desktop settings, manage taskbar options, and

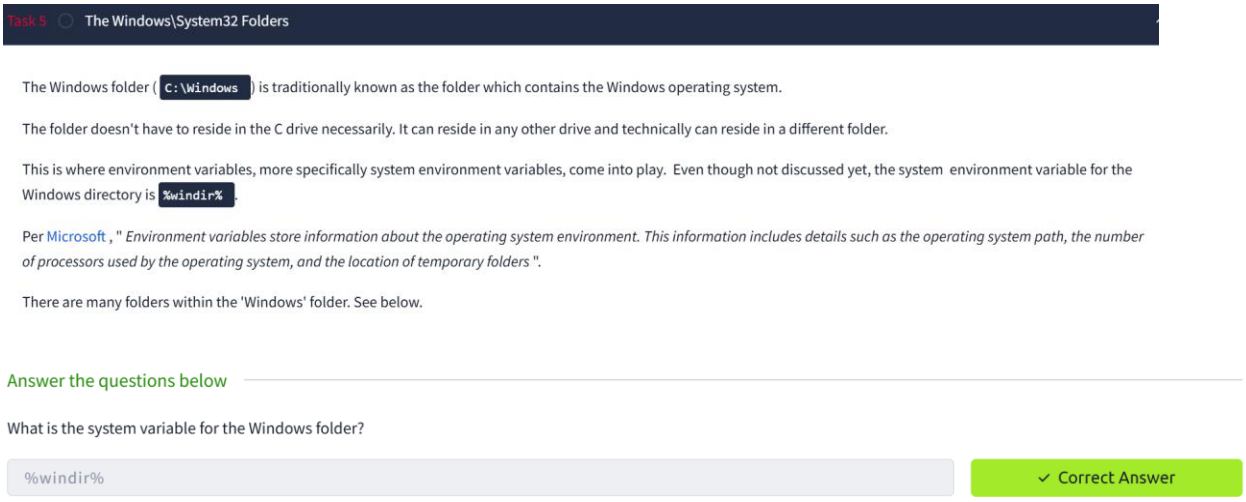
identify common system icons. This task improved my understanding of Windows navigation and basic system configuration.



Description: This section introduced the file systems used by Windows, including NTFS (New Technology File System), FAT16/FAT32, and HPFS. It explained that NTFS is commonly used in modern Windows systems and supports journaling, large files, file and folder permissions, compression, and encryption. The section also covered NTFS permissions, such as Full Control, Modify, Read & Execute, List Folder Contents, Read, and Write. These permissions help control access to files and folders. Additionally, it introduced Alternate Data Streams (ADS), a feature of NTFS that allows a file to contain multiple data streams. ADS can have legitimate uses but may also be misused to hide malicious data.

Learning Outcome:I learned the meaning of NTFS, its main features, how file and folder permissions help protect data, and how Alternate Data Streams can be relevant to cybersecurity investigations.

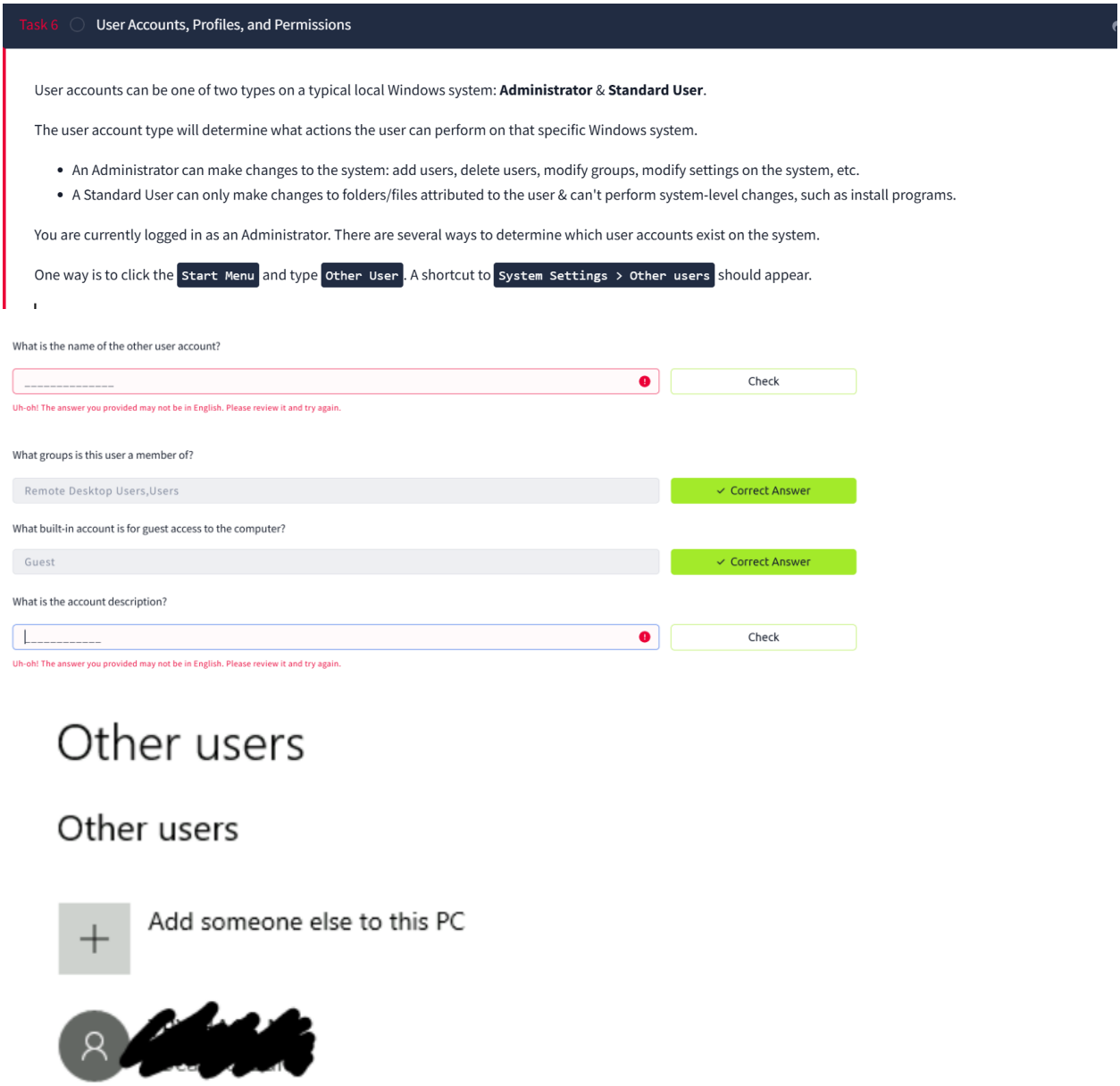
Conclusion: This task improved my understanding of Windows file systems and their role in data storage and security. I learned how NTFS supports file permissions, compression, encryption, and recovery through journaling. I also understood why Alternate Data Streams are important to know about when investigating potentially suspicious files.



Description: This section explained the Windows folder, traditionally located at C:\Windows, which contains essential files required by the Windows operating system. It introduced environment variables, which store information about the operating system environment, such as system paths and temporary folder locations. The system environment variable %windir% identifies the Windows directory, even if Windows is installed in a location other than C:\Windows.The section also introduced the System32 folder, which contains important system files and utilities required for Windows to operate correctly. Modifying or deleting files in this folder without proper knowledge can damage the operating system.

Learning Outcome:I learned the purpose of the Windows directory, the role of environment variables, and the importance of the System32 folder in maintaining the stability and functionality of Windows.

Conclusion: This task improved my understanding of the Windows directory structure and system environment variables. I learned that %windir% identifies the Windows folder and that the System32 directory contains critical operating system files. This knowledge is useful for understanding Windows administration and safely investigating system files during cybersecurity activities.



Reason for Unanswered Questions

Some answers could not be verified because the account details shown in the video lesson did not fully match the expected answer format in TryHackMe. The video was for reference only, and its contents could not be clicked to inspect account properties directly. Therefore, the exact account name and description could not be confirmed. The remaining answers were recorded based on the available lesson information.

Description: This section of the Windows Fundamentals 1 room explained the two main types of Windows accounts: Administrator and Standard User. Administrator accounts can manage system settings and user accounts, while Standard Users have limited permissions.

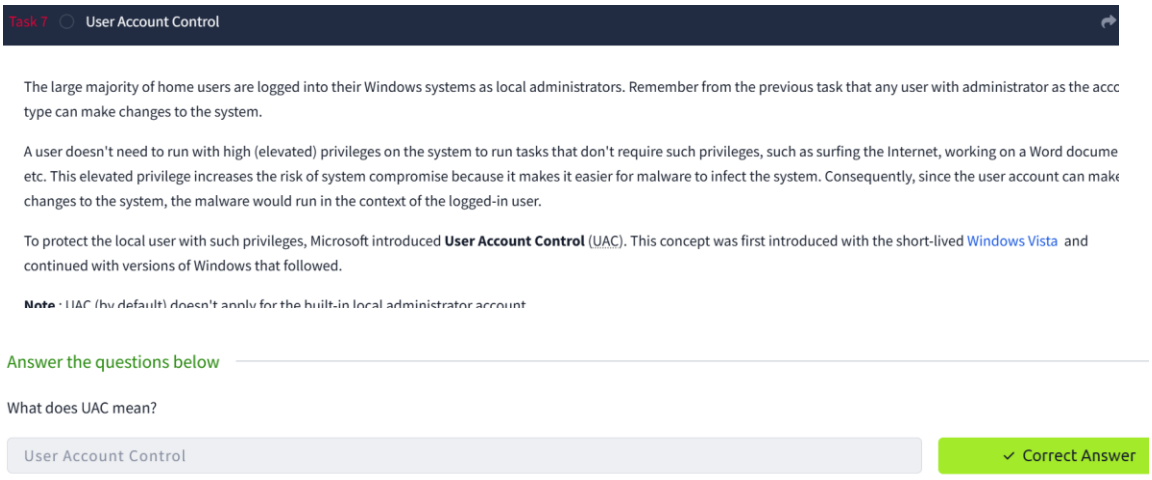
It also covered how Windows creates user profiles under C:\Users and how the Local Users and Groups Management tool (lusrmgr.msc) can be used to view user accounts, group memberships, and account descriptions.

Key Findings

- Other user account: WDAGUtilityAccount
- Group membership: Remote Desktop Users,Users
- Built-in guest account: Guest
- Guest account description: Built-in account for guest access to the computer/domain

Learning Outcome:I learned how Windows user accounts are organized, how group membership affects permissions, and how account descriptions help identify the purpose of built-in accounts.

Conclusion: This practical improved my understanding of Windows user account management, user groups, and access permissions. These concepts are important for system administration and cybersecurity because proper account management helps control access to system resources.

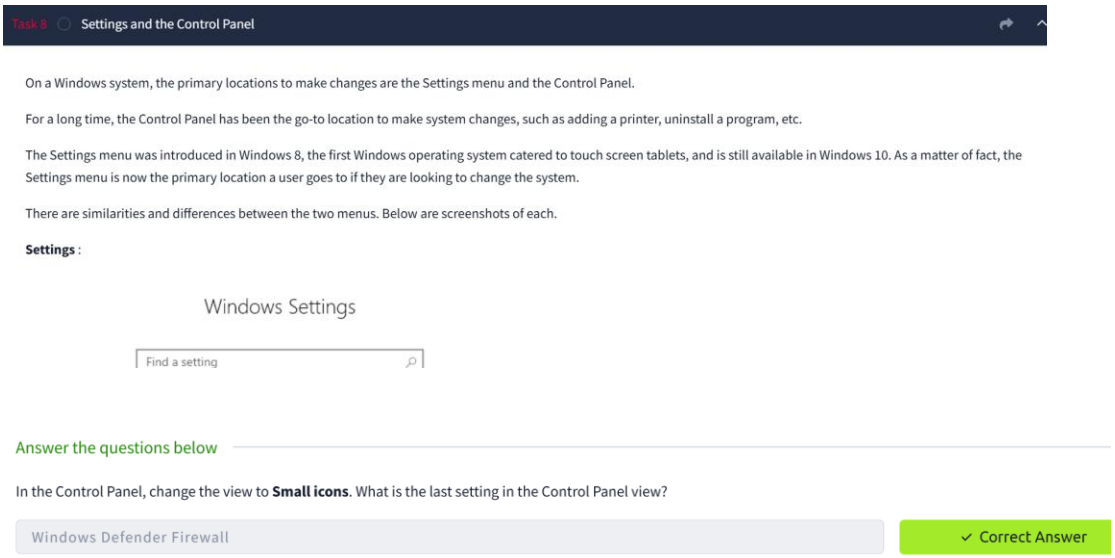


Description: This section of Windows Fundamentals 1 explained User Account Control (UAC), a Windows security feature designed to reduce the risk of unauthorized system changes. When an operation requires elevated privileges, UAC prompts the user for confirmation or administrator credentials. Running everyday applications with standard permissions helps reduce the risk of malware gaining full system access. The administrator shield icon indicates that an application may require elevated privileges to run or install. If the required credentials are not provided, the operation will not proceed.

Key Finding: UAC User Account Control

Learning Outcome:I learned how UAC helps protect Windows systems by limiting elevated privileges and requiring authorization for administrative operations.

Conclusion: This task improved my understanding of Windows privilege management and the role of UAC in system security. Using standard user accounts for everyday activities and approving administrative prompts carefully can help reduce the risk of malware infections and unauthorized system changes.

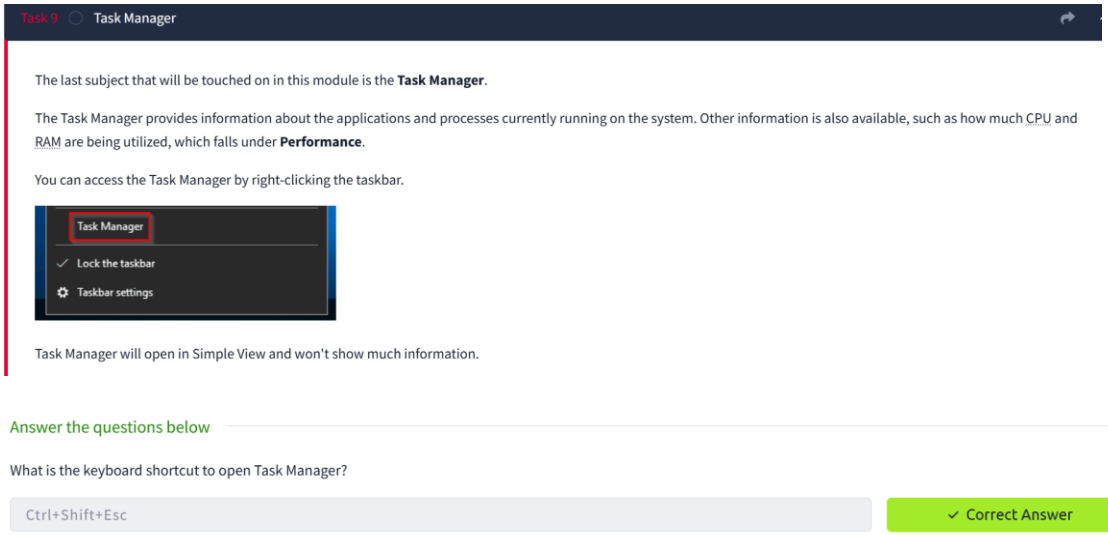


Description: This section of Windows Fundamentals 1 explained the purpose of the Windows Settings menu and Control Panel. Settings provides access to common configuration options, while Control Panel offers additional tools for managing system settings, installed applications, hardware, and network connections. The Programs and Features option allows users to view installed applications, including their names, publishers, and versions. Some settings, such as network adapter options, can be accessed through Settings and may open a Control Panel window. The Start Menu search feature also helps users locate settings quickly.

Key Finding: The last setting identified in the Control Panel's Small icons view was Windows Defender Firewall.

Learning Outcome:I learned how to navigate Settings and Control Panel, locate installed programs, and access Windows configuration tools.

Conclusion: This task improved my understanding of Windows system configuration and administration. Knowing how to use Settings and Control Panel helps users manage applications, network connections, and security settings more efficiently.



Description: This section of Windows Fundamentals 1 explained the purpose and features of Task Manager. It provides information about applications and processes currently running on a Windows system. It also displays system resource usage, including CPU and RAM utilization, through the Performance tab.

Task Manager can be opened by right-clicking the taskbar or using the keyboard shortcut Ctrl + Shift + Esc. It initially opens in Simple View, and selecting More details displays additional information about running processes and system performance.

Key Finding: Keyboard shortcut: Ctrl + Shift + Esc

Learning Outcome:I learned how to open Task Manager, view running applications and processes, and monitor CPU and RAM usage.

Conclusion: This task improved my understanding of Windows process management and system performance monitoring. Task Manager is a useful tool for troubleshooting unresponsive applications, identifying resource-intensive processes, and monitoring system activity.

